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Not feeling the buzz: Correction study of mispricing and inefficiency in online sportsbooks

Lawrence Clegg, John Cartlidge *

School of Engineering Mathematics and Technology, University of Bristol, Merchant Venturers Building, Woodland Road, Bristol, BS8 1UB, UK

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ABSTRACT

We present a replication and correction of a recent article (Ramirez et al., 2023). RRS measure profile page views on Wikipedia to generate a "buzz factor" metric for tennis players and show that it can be used to form a profitable gambling strategy by predicting bookmaker mispricing. Here, we use the same dataset as RRS to reproduce their results exactly, which confirms the robustness of RRS' mispricing claim. However, we discover that RRS' published out-of-sample betting results are significantly affected by a single bet (the "Hercog" bet), which returns substantial outlier profits based on erroneously long odds. When this data quality issue is resolved, the majority of reported profits disappear and only one strategy, which bets on "competitive" matches, remains significantly profitable in the original out-of-sample period. While one profitable strategy offers weaker support than the original study, it still provides an indication that market inefficiencies may exist, as originally claimed by RRS. As an extension, we continue testing after 2020. The strategy generates no further profits and model coefficients estimated over this period are no longer reliable predictors of bookmaker mispricing.

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1. Introduction

In a recent *International Journal of Forecasting* (IJF) article, Ramirez, Reade, and Singleton (RRS) introduced a novel "Wikipedia Relative Buzz Factor" metric (Ramirez et al., 2023).[1] This "WikiBuzz" metric, which measures excess pre-match page views of tennis players' profiles on Wikipedia, is shown to significantly predict mispricing of bookmakers' odds for tennis match outcomes. It is then used as part of a betting strategy, to test for market inefficiency. The authors report "sustained and substantial" return on investment (ROI), reaching as high as 29.38%. This is much higher than elsewhere in the tennis betting literature, where ROIs of 3.56% (Angelini et al., 2022), 3.8% (Knottenbelt et al., 2012), and 4.35% (Sipko, 2015) have been reported.

2. Replication

We present a strict replication of RRS, using the same third-party data from `tennis-data.co.uk`. We instantiate the models in new Python code, the availability of which is discussed at the end of this article.

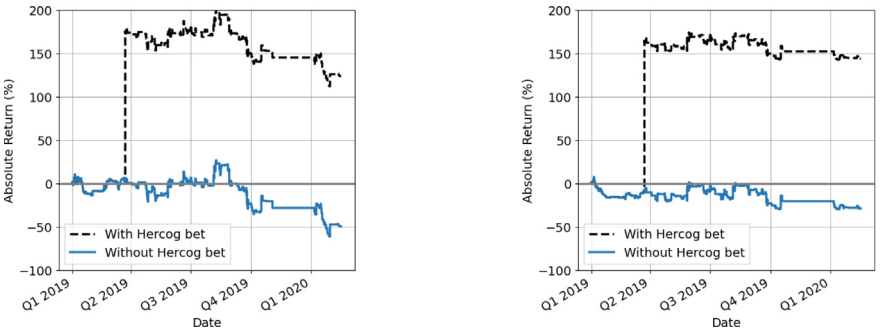
2.1. Bookmaker mispricing

RRS present results of bookmaker mispricing (Ramirez et al., 2023, Section 3.1 and Tables 1, 2, 3). Using the same data, we are able to replicate *all results* presented in these tables *exactly*. Following RRS' application of the (Mincer & Zarnowitz, 1969) forecast evaluation framework, we reach the same conclusion that "WikiBuzz" and "RankDist" coefficients are significantly non-zero. This supports RRS' finding of the existence of bookmaker mispricing.

* Corresponding author.
E-mail addresses: lawrence.clegg@bristol.ac.uk (L. Clegg), john.cartlidge@bristol.ac.uk (J. Cartlidge).

[1] The numerical results presented in the manuscript were reproduced by the Editor-in-Chief on 29 June 2024.

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(a) Preferred Model (PM) (b) Preferred Model w/o RankDist (PM w/o RD)

Fig. 1. Cumulative profits and the effect of the Hercog bet. When using Bet365 odds, the majority of reported out-of-sample returns (Ramirez et al., 2023, Table 4) are generated from a single bet on Hercog to win against Doi (March 22, 2019; see dashed line). When the Hercog bet is removed, both models make a loss (see solid line).

2.2. Market inefficiency

RRS then test market inefficiency by simulating betting strategies that utilise the bookmaker mispricing models. RRS present results of market inefficiency and betting strategies (Ramirez et al., 2023, Section 3.2 and Tables 4, 5). Again, we are able to replicate these results when using the same dataset as RRS for training and testing, and we are able to confirm that the resultant series of bets placed are identical to RRS.[2]

However, for the best-performing betting models, we find that a large proportion of profits are generated by a single bet (the "Hercog" bet) on Bet365, which results from an inconsistency (i.e., an *error*) in the odds data provided by `tennis-data.co.uk`.[3] In Fig. 1, we plot the cumulative profit and loss graphs for two of the most profitable betting models proposed in the original paper (Ramirez et al., 2023, Table 4, Column (III) and Column (IV)). The first, which we refer to as the "preferred model" (PM), has reported returns of 17.3% when using Bet365 odds; and the second, which is a variation of the preferred model *without* the rank distance to opponent metric (PM w/o RD), has reported returns of 28.8% on Bet365 odds.

The sudden surge in cumulative returns to over 150% is due to a single large bet (of size 39.3% bankroll) on Polona Hercog to win against Misaki Doi in the 2019 Miami Open. Despite the data recording the highest odds on Hercog to win (collected from a sample of between 40–60 bookmakers) as 2.10, and the market's average odds as 1.85, the Bet365 odds are listed as 5.50. This is a clear contradiction: the Bet365 odds on Hercog to win are an error. Presented with these extremely long odds, the Kelly criterion anticipates significant bookmaker mispricing and specifies a proportionately large stake, which results in abnormal returns when Hercog wins.[4] This affects all Bet365 betting results presented by RRS, and requires a correction.

3. Corrections: Market inefficiency results

Table 1 shows corrected results of the most profitable models presented in RRS, Table 4. RRS shows the original published values; CC shows replication results on the original dataset (including the Hercog bet); and CC/H shows corrected results when the Hercog row is removed from the dataset. We see that strict replication results (CC) are the same as the original (RRS).[5] However, when the Hercog bet is removed (CC/H), the exceptional ROIs of 17.29% and 28.82% for PM and PM w/o RD, respectively, fall to negative values (i.e., *losses*) of −7.36% and −6.31%. Consequently, final absolute returns are negative (−50% and −29%), as shown in the cumulative profits plotted in Fig. 1.

Table 2 shows corrected results of the models presented in RRS, Table 5. These betting strategies explore the effects of match competitiveness by only sampling matches (for model estimation and betting) with average odds implied probability p within some interval range. When $p = 0.5$, the market's average odds suggest an even match, with no favourite. As p approaches 0 or 1, there is a clear favourite. For Hercog, $p = 0.54$; i.e., this is a highly "competitive" match and Hercog is only a very slight favourite over Doi (despite the erroneous Bet365 odds of 5.5 listed on `tennis-data.co.uk`). Therefore, RRS, Table 5, Columns III (PM w/o RD: $p \in [0.2, 0.8]$) and IV (PM w/o RD: $p \in [0.4, 0.6]$) are affected by the Hercog bet. In

[2] Link to RRS' code: https://github.com/philiprami/betting_on_a_buzz. Link to RRS' data: https://drive.google.com/drive/folders/1GiRMkek1MnUAYzIhutt_Oie_Xlq2vDO7.

[3] RRS' dataset contains WTA data sourced from `tennis-data.co.uk` for the period July 2015 to February 2020, inclusive; which includes information on every tennis match (competitors, competitor rankings, tournament, location, date, result, etc.) and a selection of pre-match betting odds taken from the major bookmakers, such as Bet365, as well as the market's average odds and best odds taken from the K individual bookmakers listed on `oddsportal.com` (where K is normally in the range 40–60). A full description of the tennis data is available online: tennis-data.co.uk/notes.txt. For Hercog to beat Doi, `oddsportal.com` reportsoddsof2.10. Therefore, the Hercog error likely originates from `tennis-data.co.uk`.

[4] In comparison, a unit betting strategy (i.e., place £1 on every mispriced bet) would exhibit a much less pronounced spike in returns from the Hercog error.

[5] The only difference is ROI for PM. We have confirmed that the RRS reported value of 17.26 is a simple typo, as RRS' original Stata code also indicates ROI value of 17.29.

Table 1
Corrected out-of-sample Bet365 strategy results for WTA, 2019–2020.

	Bet365 Odds					
	PM			PM w/o RD		
	RRS	*CC*	*CC/H*	*RRS*	*CC*	*CC/H*
N odds ($2 \times J$ matches)	5156	5156	5155	5156	5156	5155
Number of bets placed	312	312	311	276	276	275
Mean overround (%)	6.46	6.46	6.46	6.46	6.46	6.46
Investment (×per bet budget)	7.15	7.15	6.77	4.99	4.99	4.60
Absolute return (×per bet budget)	1.24	1.24	−0.50	1.44	1.44	−0.29
Return on Investment (%)	17.26	17.29	−7.36	28.82	28.82	−6.31

1. Notes: RRS shows results copied from original paper: PM (Ramirez et al., 2023, Table 4, Column III); PM w/o RD (Ramirez et al., 2023, Table 4, Column IV). CC: Strict replication. CC/H: Replication without Hercog bet.
2. Replication: CC demonstrates an exact replication, apart from the ROI of PM. The original value 17.26 is a typo.
3. Hercog bet: CC/H shows the impact of removing the single Hercog bet. ROI is negative for both models.

Table 2
Corrected out-of-sample Bet365 strategy results for WTA, 2019–2020: selecting sample average bookmaker odds, p, based on match competitiveness.

	Bet365 Odds					
	PM w/o RD: $p \in [0.2, 0.8]$			PM w/o RD: $p \in [0.4, 0.6]$		
	RRS	*CC*	*CC/H*	*RRS*	*CC*	*CC/H*
N odds ($2 \times J$ matches)	4424	4454	4453	1697	1697	1696
Number of bets placed	363	384	383	263	263	262
Mean overround (%)	6.58	6.58	6.58	7.01	7.01	7.01
Investment (×per bet budget)	7.25	7.95	7.57	9.27	9.27	8.91
Absolute return (×per bet budget)	1.46	1.44	−0.25	2.72	2.72	1.11
Return on Investment (%)	20.11	18.13	−3.36	29.38	29.38	12.44

1. Notes: RRS shows results of PM w/o RD copied from original: $p \in [0.2, 0.8]$ (Ramirez et al., 2023, Table 5, Col. III); $p \in [0.4, 0.6]$ (Ramirez et al., 2023, Table 5, Col. IV). CC: Replication. CC/H: Replication without Hercog bet.
2. Replication: CC demonstrates a close replication. Minor differences in results for PM w/o RD are caused by a floating point error at the upper boundary $p = 0.8$, which causes 30 potential bets to be overlooked in the original.
3. Hercog bet: CC/H shows the impact of removing the single Hercog bet. Strategy $p \in [0.2, 0.8]$ has negative ROI. However, for strategy $p \in [0.4, 0.6]$, ROI falls but remains significantly positive.

Table 2, CC shows replicated results, including the Hercog bet. Note that, while replication results (CC) are close to the original (RRS), there are some minor differences due to RRS's Stata code containing a floating point precision error for value $p = 0.8$.[6] As a result, our replication places an additional 21 bets and produces a slightly lower ROI than RRS. Much more significantly, when the Hercog bet is removed (CC/H), the strategy that samples $p \in [0.2, 0.8]$ makes a loss. However, for $p \in [0.4, 0.6]$, i.e., when only the most competitive matches are sampled for training and testing, ROI remains positive, albeit at a much lower rate of 12.44% (compare with previously reported value of 29.38%, as shown in column CC).

To assess the informative value of the 262 bets placed by the remaining profitable strategy, we refer to the betting return significance framework proposed by Wunderlich and Memmert (2020) and calculate the probability p_{bs} that a random betting strategy would have produced the same or a higher return as this positive ROI strategy. Here, we find $p_{bs} = 0.002$; i.e., in 100,000 repeated simulation trials of randomly placing 262 bets in our dataset, 0.2% of all random strategies generated equal or higher returns. Therefore, after Hercog error correction, we confirm that a profitable Bet365 strategy exists and it is unlikely to be a result of chance alone. While we note that the possibility of errors in the Wikipedia page view data could impact the reliability of this result, these consistent profits provide an indication of the existence of market inefficiencies.

4. Extension

We extend the original work of RRS by exploring the robustness of strategy PM w/o RD for competitive matches $p \in [0.4, 0.6]$ on Bet365 odds over an extended period; and we explore bookmaker mispricing over this extended period using the preferred model (PM). We first clean the data of outliers.[7]

4.1. Market inefficiency

We tested the strategy (using the original model coefficients) over an extended out-of-sample period: Feb. 2020 – Aug. 2023.

However, we find that no further profits are gained over this period, potentially suggesting that markets have

[6] 120 odds (90 in-sample; 30 out-of-sample) that should be $p = 0.8$ are incorrectly stored by RRS as 0.800000011920929. We include these odds for strategy $p \in [0.2, 0.8]$.

[7] We first drop rows where Bet365 odds are higher than the best odds. We then drop rows with any missing values. Finally, we drop rows with suspiciously high best odds, such that the absolute difference between the best odds in the market and average odds of the market is more than four times the standard deviation of the mean difference between best odds and average odds.

become more efficient, although other unknown factors may contribute to this apparent loss of profitability.

4.2. Bookmaker mispricing

We then tested bookmaker mispricing by replicating the original mispricing test (Ramirez et al., 2023, Table 1, Column IV) on the preferred model (PM) over the period Feb. 2020 – Aug. 2023. Over 13,913 potential bets (i.e., N odds) during this period, we find that estimated coefficients for WikiBuzz and RankDist are no longer significant, with respective p-values of 0.84 and 0.47. We present this as tentative evidence that, since 2020, WikiBuzz and RankDist factors are no longer reliable predictors of bookmaker mispricing.

5. Conclusion

We have replicated and corrected an error in the work of Ramirez et al. (2023). While the correction significantly affects reported betting returns, when considering the original data period our replication findings do not contradict the original authors' claims of bookmaker mispricing and market inefficiencies. In particular, our finding of significant profitability for the Bet365 selective strategy based on "competitive" matches (Section 3) may indicate the existence of market inefficiencies. However, the introduction of a new data period (Section 4) resulted in the loss of the model's predictive power for both inefficiency and mispricing, thus highlighting the need for further research to account for potential limitations of the model.

Our work exemplifies the sensitivity of betting models to data quality, and expands upon the findings of Wunderlich and Memmert (2020) to demonstrate that even a *single* data error can significantly influence ROI when adopting a Kelly staking strategy. Therefore, simulated betting returns should be interpreted cautiously, and data quality should always be checked, even when using popular open source datasets such as `tennis-data.co.uk`.

More generally, our findings also demonstrate the importance of replication studies in forecasting research. In this instance, having access to the original data has proven invaluable and we thank the authors for making their data and code available. The *International Journal of Forecasting* (IJF) – the venue where the original work was published – promotes replication (Hyndman, 2010) and from July 2023 has made data and code sharing mandatory. IJF also publishes replication studies (e.g., Boylan et al., 2015). However, as described at length by Makridakis et al. (2018), not all journals take such an objective standpoint on replication and reproducibility. We, the authors, strongly agree with the imperative of replication and to help address this issue, one of us (Cartlidge) encourages the use of replication studies when supervising undergraduate and postgraduate student projects. To this end, the preliminary research that formed the basis of this present paper was initially performed by Clegg as part of his MSc project in Financial Technology with Data Science at the University of Bristol. In previous years, students supervised by Cartlidge have had success publishing replication studies that reproduce and extend previous work (e.g., Duffin & Cartlidge, 2018), or provide some corrections to previous work (e.g., Cartlidge & Clamp, 2014; Clamp & Cartlidge, 2013; Stotter et al., 2013, 2014). Harnessing student research hours for replication studies is beneficial for all parties, but encouraging students to perform replication is challenging if their results cannot be published. Therefore, we commend the IJF's approach to reproducible research and we urge other journal venues to follow suit, if they do not already.

An extended version of this paper, with full analysis, is available online.[8]

CRediT authorship contribution statement

Lawrence Clegg: Conceptualization, Data curation, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing, Formal analysis. **John Cartlidge:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The code and data necessary to reproduce the work presented in this article is made available at https://github.com/Faxulous/notFeelingTheBuzz.

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